

EcoCast Database Schema Documentation

Project Overview

EcoCast is an automated ETL pipeline that extracts weather forecast data from the Open-Meteo API, transforms the JSON response into structured tabular data, and stores it in PostgreSQL for renewable energy analytics and dashboard visualization.

The database is designed using a relational schema with primary and foreign key relationships to support scalable querying and analytics workflows.

Table 1: Locations

	Data type	Key	Description
Location_id	Serial	Primary key	Unique identifier for each location
City_name	Varchar(100)		Name of the city
Latitude	Decimal(8,5)		Geographic latitude
Longitude	Decimal(8,5)		Geographic longitude
timezone	Varchar(100)		Timezone of the location

Table 2: weather_forecasts

	Data Type	Key	Description
forecast_id	SERIAL	Primary key	Unique forecast record identifier
location_id	INTEGER	Foreign key	References locations.location_id
forecast_time	TIMESTAMP		Forecast date and time
temperature	DECIMAL(5,2)		Temperature in Celsius
cloud_cover	INTEGER		Percentage cloud coverage
wind_speed	DECIMAL(5,2)		Wind speed in km/h
wind_direction	INTEGER		Wind direction in degrees

Table 3: renewable_scores

	Data Type	Key	Description
score_id	SERIAL	Primary Key	Unique renewable score identifier
forecast_id	INTEGER	Foreign Key	References weather_forecasts.forecast_id
solar_score	DECIMAL(5,2)		Estimated solar generation potential
wind_score	DECIMAL(5,2)		Estimated wind generation potential
recommendation	VARCHAR(100)		Suggested renewable energy condition

Database Relationships

- The EcoCast database follows a relational structure that connects locations, weather forecasts, and renewable energy analytics.
- Each location may contain multiple weather forecast records.
- Each weather forecast record may contain one renewable energy scoring record.
- Foreign key constraints are used to maintain referential integrity across all related tables.

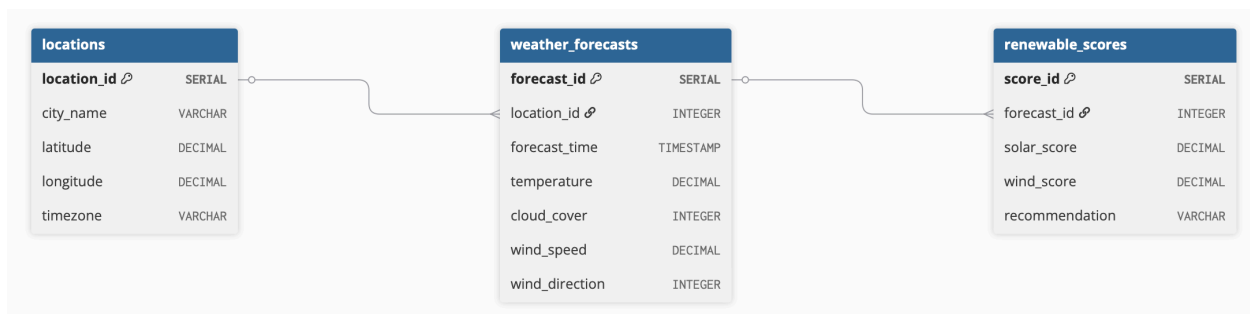
Relationship Structure:

locations (1) ---- (Many) weather_forecasts

weather_forecasts (1) ---- (1) renewable_scores

ERD

Figure 1. Entity Relationship Diagram (ERD) illustrating the relational structure of the EcoCast PostgreSQL database schema.



Initial PostgreSQL Load Script

```
CREATE TABLE locations (  
    location_id SERIAL PRIMARY KEY,  
    city_name VARCHAR(100) NOT NULL,  
    latitude DECIMAL(8,5),  
    longitude DECIMAL(8,5),  
    timezone VARCHAR(100)  
);  
  
CREATE TABLE weather_forecasts (  
    forecast_id SERIAL PRIMARY KEY,  
    location_id INTEGER NOT NULL,  
    forecast_time TIMESTAMP NOT NULL,  
    temperature DECIMAL(5,2),  
    cloud_cover INTEGER,  
    wind_speed DECIMAL(5,2),  
    wind_direction INTEGER,  
    CONSTRAINT fk_location  
        FOREIGN KEY (location_id)  
        REFERENCES locations(location_id)  
);  
  
CREATE TABLE renewable_scores (  
    score_id SERIAL PRIMARY KEY,  
    forecast_id INTEGER NOT NULL,  
    solar_score DECIMAL(5,2),  
    wind_score DECIMAL(5,2),  
    recommendation VARCHAR(100),  
    CONSTRAINT fk_forecast  
        FOREIGN KEY (forecast_id)  
        REFERENCES weather_forecasts(forecast_id)  
);
```

The PostgreSQL schema creation script was successfully executed in the SQL editor environment without errors. All tables, primary keys, and foreign key relationships were successfully created.

Figure 2. Successful execution of the PostgreSQL schema creation script.

```
22 CREATE TABLE renewable_scores (  
23     score_id SERIAL PRIMARY KEY,  
24     forecast_id INTEGER NOT NULL,  
25     solar_score DECIMAL(5,2),  
26     wind_score DECIMAL(5,2),  
27     recommendation VARCHAR(100),  
28     CONSTRAINT fk_forecast  
29         FOREIGN KEY (forecast_id)  
30         REFERENCES weather_forecasts(forecast_id)  
31 )
```

Results Explain Chart Export ▾

Source Primary Database ▾ Role postgres ▾ Run ⌘+R

Success. No rows returned

The schema is designed to support future ETL automation, analytical querying, and renewable energy dashboard reporting.